

The Capstone Submission — Paper · Replication Packet · Presentation

NextGen (NGN) Empirical Finance Research Camp · Prof. Lei Gao (George Mason University) Week 8 (Independent Research Project II) — assigned Fri Aug 14, 2026 · **DUE Friday, August 15, 2026 (presentation/conference day)**. This is the program's **terminal deliverable** — the paper you have been building for two weeks.

— Why this assignment

Everything in eight weeks was preparation for this. You learned to estimate, to identify, to read the frontier, to use AI without being fooled, to design a study, and to pre-register it. Now you **execute the plan you filed, break your own result on purpose, write it to publication standard, and defend it**. The deliverable is not just a set of results — it is a complete, reproducible, honestly-argued research paper plus the packet that lets anyone verify it and the talk that lets you defend it.

One rule above all others: report the confirmatory result first and whatever it is — effect or null — and let your **design**, not your p-value, carry the paper. A clean null, honestly reported, is worth more than a "significant" result fished from the multiverse.

— Your task — four pieces

Part 1 — The paper (the core)

Write the empirical paper. Required structure:

- **Abstract** — **≤ 250 words**: question, design, data, headline finding, one-sentence contribution.
- **Introduction** — the five-sentence skeleton expanded: what question, why it matters, what you do, what you find, what it contributes.
- **Data & design** — the identifying assumption and threats table from your Week-7 memo, now with the data actually built.
- **Results** — the **confirmatory** primary specification **first**, reported whatever it says; then the **specification curve** (where does your headline sit in the multiverse?) and the **robustness battery** (placebos, alternative SEs, wild-cluster bootstrap, Oster δ).
- **Discussion & limits** — what the design does and does not license; the honest limitations.
- **A deviations log** — every departure from your pre-analysis plan, disclosed.

Causal-language discipline: match the verb to the design. "Associated with" for observational; "causes/raises/lowers" only where identification earns it.

Part 2 — The one-click replication packet

A repository that **regenerates every number** in your paper from raw data with **one command** (a `Makefile` or `run_all` script). Include: pinned environment, seeded code, the data-pull with a pinned vintage, and a `README` that says exactly how to run it. If a reviewer

Part 3 — The 8-minute presentation

An 8-slide (roughly) deck, one idea per slide: **1** motivation · **2** design/identification · **2** results · **1** robustness · **1** contribution · **1** limits. Rehearse to time. Prepare the **three honest answers** for Q&A: "I don't know — here's how I'd find out," "that's a limitation — here's how I bounded it," "here's the exact evidence."

Part 4 — The AI-use disclosure

One short paragraph: where and how you used AI in this project (triage, code, prose), and how you **verified** what it gave you. Honesty here is graded; hidden use is not acceptable.

— What to hand in

1. The **paper** (PDF) — abstract, intro, data & design, results (confirmatory + spec curve + robustness), discussion, deviations log. 2. The **replication packet** (repo link or archive) — one-command reproducibility. 3. The **8-minute presentation** (slides) — ready to deliver on Aug 15. 4. The **AI-use disclosure** (one paragraph, in the paper or as a note).

Due: Friday, August 15, 2026. Bring the talk ready to give.

— How you'll be graded — the Capstone rubric (200 points)

- **Design & identification (50).** A clear identifying-assumption sentence and threats table; the verb matches the design; limitations are named, not hidden.
- **Execution & robustness (50).** The confirmatory spec is reported first; a real specification curve; a robustness battery with tests that could have failed; a deviations log.
- **Writing (40).** A ≤ 250 -word abstract; the five-sentence spine is legible; a true, supported one-sentence contribution; disciplined causal language.
- **Reproducibility (30).** One command regenerates every number; pinned env + vintage; seeded; a reviewer could rerun it.
- **Presentation & defense (20).** An 8-minute talk, one idea per slide; the three honest Q&A answers; you can defend your identification out loud.
- **AI-use disclosure (10).** Present, specific, and honest about verification.

You are doing great if a skeptic reading your paper cannot find a claim your design does not support, and a stranger cloning your repo reproduces your headline number on the first try.

— Tips

- **Report the confirmatory result first — always.** Then explore. The order is the evidence you did not fish.
- **Put your headline on the specification curve.** "Our estimate is the black dot; it keeps its sign in 130 of 144 specifications" is one of the most persuasive sentences you can write.
- **Design a robustness test that could fail.** A placebo that might have lit up and didn't is worth a paragraph of prose.
- **Cut the abstract to 250 words.** If it does not fit, your contribution is not sharp enough yet.

Discipline every causal verb. The most common capstone error is an association dressed as causation. Read your results section once looking only at verbs.

- **Make the repo run for a stranger.** Clone it into a fresh folder and run the one command yourself before you submit.
- **Rehearse the talk to time, out loud, once.** Eight minutes is short; the only way to know is to run it.

You built this over eight weeks. Now go defend it. See you at the presentation — **August 15.**